

THE MURMURATION OF A CM QUADRATIC TWIST FAMILY: EXACT REDUCTION, CLOSED-FORM ROOT NUMBERS, THE MEASURED GRÖSSENCHARACTER WAVEFORM, AND THE VERIFIED POISSON ENGINE

ABSTRACT. For the quadratic twist family of $E_0 = 27a3$ ($y^2 + y = x^3$, CM by $\mathbb{Q}(\sqrt{-3})$) we prove an exact reduction of the murmuration to elementary arithmetic: the root numbers of the entire family admit a two-line closed form (via a CM self-twist argument eliminating the local analysis at 3; verified 153/153), the traces are given by $4p = L^2 + 27M^2$, and the density factors through a single character sum over fundamental discriminants. The resulting pipeline is certified against exact computation to five decimal places and requires no elliptic-curve arithmetic. We then reduce the family waveform to a single-object function: the murmuration $\overline{M}(s)$, $s = p/N_D$, of the fixed Hecke Grössencharacter of $\mathbb{Q}(\sqrt{-3})$ under quadratic twist. Empirically $\overline{M}$ is universal across disjoint discriminant ranges (correlation +0.945) and is resolved at $5\text{--}16\sigma$: a principal peak +7.4 near $s \approx 0.03$, oscillation with a deep trough -8.3 just below the conductor scale and a rebound +7.3 just above it, decaying to the explicit-formula constant $C_0 = \frac{1}{2}\overline{w} - \overline{w}r = +0.4085$ (confirmed exactly from rank parities). The exact transform of $\overline{M}$ reproduces the directly measured family profile on its resolved bins. The derivation of $\overline{M}$ in closed form is located within the published murmuration framework: the Dirichlet-character (GL(1)) densities of Lee–Oliver–Pozdnyakov, the explicit-formula mechanism for quadratic families, and Sarnak’s conductor-growth framework; our object—the quadratic twist family of a Grössencharacter over an imaginary quadratic field—appears to be new, and every arithmetic ingredient of its Zubrilina-type computation is supplied here in verified closed form.

1. THE EXACT REDUCTION

Let $E_D = E_0^{(D)}$ for fundamental discriminants D , and $F(X) = \{0 < |D| \leq X\}$. All results below are machine-certified; scripts are available with this note.

Proposition 1 (Reduction). *For p of good reduction for the family, $\frac{1}{|F|} \sum_D w(E_D) \frac{a_p(E_D)}{\sqrt{p}} = \frac{a_p(E_0)}{\sqrt{p}} S(p)$ with $S(p) = \frac{1}{|F|} \sum_D w(E_D) \left(\frac{D}{p}\right)$. Verified over 57,681 pairs; the two sides, computed independently, agree to 6.9×10^{-18} .*

Proposition 2 (Closed-form root numbers). *For fundamental D , $|D| \leq 250$: $w(E_D) = \left(\frac{D}{-27}\right)$ if $\gcd(D, 3) = 1$, and $w(E_D) = \left(\frac{-D/3}{-27}\right)$ if $3 \mid D$. Proof for $3 \mid D$: the self-twist $E_0^{(-3)}$ has $a_p(E_0^{(-3)}) = \chi_{-3}(p)a_p(E_0) = a_p(E_0)$ at every good prime, additive reduction at 3 on both sides, and equal conductor; hence identical L -functions and equal functional-equation sign, so with $D = (-3)m$ (prime-discriminant factorization), $w(E_D) = w(E_0^{(m)})$, reducing to the coprime case. Verified 114/114 (coprime, including even D) and 39/39 ($3 \mid D$).*

Corollary 3 (Curve-free computability). *With $a_p(E_0) = 0$ at inert p and, at split p , the root of $4p = L^2 + 27M^2$ congruent to 2 mod 3, the murmuration is elementary arithmetic. The pipeline reproduces exact computation to five decimals ($+0.03011 = +0.03011$), and $S(p)$ factors through $T(Y; p) = \sum_{\text{fund } m, 3 \nmid m, |m| \leq Y} \left(\frac{m}{-27p}\right)$.*

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2. REDUCTION TO A SINGLE GRÖSSENCHARACTER; THE MEASURED WAVEFORM

Every E_D has CM by $\mathbb{Q}(\sqrt{-3})$: its L -function is that of a Hecke Grössencharacter. Define the *single-object murmuration*

$$\overline{M}(s) = \langle w(E_D) a_p(E_D) \rangle_{p=sN_D}, \quad N_D = \begin{cases} 27D^2, & 3 \nmid D, \\ 3D^2, & 3 \mid D, \end{cases}$$

each twist in its own conductor-scaled variable. The family profile is the exact transform $g(x) = \langle \overline{M}(x \cdot 27X^2/N_D) \rangle_{D \in F(X)}$.

Observation 4 (Universality and the waveform). Superposing all 153 twists (1,370,136 samples, inverse-variance weighted), $\overline{M}$ measured from $|D| \leq 100$ and from $|D| > 100$ agree with correlation +0.945: the single-object waveform exists. Its measured values:

s -bin	$\overline{M}(s) \pm \text{SE}$	feature
[0, 0.02)	+0.01 $\pm$ 0.13	flat onset
[0.02, 0.05)	+7.40 $\pm$ 0.46	principal peak
[0.05, 0.10)	+4.10 $\pm$ 0.57	
[0.10, 0.20)	−4.17 $\pm$ 0.60	first sign change
[0.20, 0.35)	−1.61 $\pm$ 0.72	
[0.35, 0.50)	+5.60 $\pm$ 0.97	second peak
[0.50, 0.75)	+0.20 $\pm$ 1.02	
[0.75, 1.00)	−8.31 $\pm$ 1.26	pre-conductor trough
[1.00, 1.50)	+7.33 $\pm$ 1.12	post-conductor rebound
[1.50, 2.50)	+0.01 $\pm$ 1.10	
[2.50, 4.00)	+1.41 $\pm$ 1.32	decay toward C_0

The structure at the conductor scale (trough below $s = 1$, rebound above) is the functional-equation transition made visible. Reproduced deterministically on independent hardware from an 80-line script using only the closed forms.

Observation 5 (Transform closure). The measured $\overline{M}$, pushed through the exact transform, predicts the directly measured family profile: on its two resolved bins, predicted +1.79, +1.91 versus measured +2.19 $\pm$ 0.42, +1.98 $\pm$ 0.57; 13 of 15 bins agree within 1σ . The apparent irresolution of the family profile beyond its front is thereby explained as *phase-mixing*: each twist oscillates at its own conductor scale, and the family variable averages $\overline{M}$ over a wide range of its argument.

Observation 6 (The constant). $C_0 = \frac{1}{2}\overline{w} - \overline{w\overline{r}}$ with exact ranks gives +0.4085 ($\overline{w} = -7/153$, $\overline{w\overline{r}} = -66/153$), derived from rank parities in advance and reproduced exactly; $C_0 \rightarrow \frac{1}{2}$ under Goldfeld and parity.

3. THE POISSON ENGINE, VERIFIED

The remaining computation is now implemented and machine-verified. The Hecke character of E_0 has the exact lattice description $\psi_0(\mathfrak{a}) = \lambda$, the unique generator with $\lambda \equiv 1 \pmod{3}$; equivalently, for $\gcd(n, 3) = 1$,

$$a_n(E_0) = \sum_{\substack{a \equiv 1, \, b \equiv 0 \pmod{3} \\ a^2 - ab + b^2 = n}} (a + b\omega),$$

verified exactly for all $n \leq 2000$ (zero mismatches), and the twisted sums $\sum_n \chi_D(n) a_n g(n/P)$ equal the corresponding lattice sums to machine precision. Two-dimensional Poisson summation over $\mathbb{Z}[\omega]$, with the Gaussian-times-linear envelope transformed in closed form (checked against numerical integration at 10^{-9}), then gives, for $P = sN_D$,

$$\sum_n \chi_D(n) a_n(E_0) e^{-2\pi n/(sN_D)} = s^2 \sum_{\mu \in \mathbb{Z}^2} \gamma_\mu(D) e^{-2\pi s Q^*(\mu)}, \quad Q^*(\mu) = \mu_1^2 + \mu_1 \mu_2 + \mu_2^2,$$

with $\gamma_\mu(D)$ explicit finite Gauss sums (the discrete Fourier coefficients of the periodic datum: primarity mod 3 and $\chi_D \circ Q \bmod |D|$). The identity is verified at machine precision (1.3×10^{-14} at $s = 1$; 3.5×10^{-15} at $s = 0.05$) for $3 \nmid D$ ($D = 5$, $M = 3|D|$) and for $3 \mid D$ ($D = -24$, $M = |D|$); in both cases the conductor normalization makes the dual exponent *exactly* $2\pi s Q^*(\mu)$ —deriving, rather than assuming, that the murmuration variable is $s = p/N$. The dual support is sharply localized on hexagonal shells: at $s = 1$ the first shell ($Q^* = 1$, six vectors) carries 100% of the sum; three shells at $s = 0.3$; ten at $s = 0.05$ —an $O(1/s)$ term count matching the measured waveform’s simple conductor-scale structure and rich front. The truncation $Q^* \leq 30$ reproduces the exact sum at relative error 10^{-15} for $s \geq 0.1$.

Step (i) is now closed in the generic case. For odd fundamental D with $3 \nmid D$ (period $M = 3|D|$, $\alpha \equiv |D|^{-1} \bmod 3$):

$$\gamma_\mu(D) = \frac{1}{9|D|} \omega_3^{\alpha\mu_1} \chi_{-3}(|D|) \chi_D(-3) \chi_D(Q^*(\mu)),$$

derived by completing the square against the norm form (whose Gauss sum is $\chi_{-3}(q)q$, independent of the multiplier—the hexagonal miracle) with the $\tau(\chi_D)$ factors cancelling identically. Verified at 10^{-17} against brute discrete transforms across six discriminants (224 dual vectors each), and the resulting fully explicit dual formula—kronecker symbols and exponentials only—reproduces the exact lattice sums at 10^{-11} – 10^{-13} . Since $w(E_D) = \chi_D(-27) = \chi_D(-3)$ for these D , the root-number weight cancels the Gauss phase exactly:

$$w_D \gamma_\mu(D) = \frac{\chi_{-3}(|D|)}{9|D|} \omega_3^{\alpha\mu_1} \chi_D(Q^*(\mu)) :$$

the murmuration weight is precisely the conjugating phase of the dual expansion—the structural reason this family murmurs coherently. For the excluded cases the same skeleton holds numerically: $|\gamma| \cdot 9|D| = 1$ exactly for even D and $= 3$ for $3 \mid D$ (ramification enhancement), with the $\chi_D(Q^*)$ -proportionality exact in all tested vectors; only the unit phases at 2 and 3 remain as classical case-work.

Step (ii) is now closed at the main-term level for the generic subfamily. The character sums $T_\mu(X) = \sum_D D^2 \chi_{-3}(|D|) \omega_3^{\alpha\mu_1} \chi_D(Q^*(\mu))$ have main terms exactly at square shells $Q^*(\mu) = k^2$ with $\mu_1 \not\equiv 0 \bmod 3$ (verified: alive vectors converge at X^3 -order; dead vectors, dead shells such as $Q^* = 9$, and non-square shells decay — within the single shell $Q^* = 1$, $\mu = (1, 0)$ converges while $\mu = (0, 1)$ dies). Pairing each shell in $\pm\mu$, the real part of T_μ cancels identically—the equidistribution error drops out exactly—and the shell geometry evaluates to $c^T A_0^{-1}(2k, -k) = 2k$, giving the closed-form main term of the w -averaged Gaussian-smoothed family sum:

$$\lim_{X \rightarrow \infty} \frac{A(X, s)}{\sum_{|D| \leq X} D^2} = 27 s^2 \sum_{k \geq 1} \chi_{-3}(k) k e^{-2\pi s k^2}$$

(up to Euler-factor corrections for k with prime factors ≥ 5 and extra Lösschian representations from $k^2 = 49$ onward, both entering at $e^{-50\pi s}$). The formula reproduces the computed family averages to all printed digits at $s = 0.5$, 1 ($+0.29165$, $+0.05042$), and at small s the non-square residual is exhibited decaying in X ($-5.7 \times 10^{-4} \rightarrow -6.0 \times 10^{-5}$ over $X = 400 \rightarrow 3200$ at

$s = 0.05$). The main term is a harmonic theta series with character χ_{-3} —an η^3 -adjacent object whose modular identification we flag but do not assert.

Step (iii), part one—the unsmoothing—is closed and confirmed against data. Laplace inversion of the theta main term (the classical pair $\mathcal{L}\{\sqrt{u/a} J_1(2\sqrt{au})\} = \lambda^{-2} e^{-a/\lambda}$) gives the coefficient-level density in closed form,

$$\rho(u) = 2\pi\sqrt{u} \sum_{k \geq 1} \frac{c_k}{27k} J_1(4\pi k\sqrt{u}),$$

a Fourier–Bessel (Voronoi-kernel) series with the shell coefficients computed by exact enumeration: $c_1 = 27$, $c_2 = -54$, $c_4 = 108$, $c_5 = -112.5$, $c_7 = -141.75$ (sign flipped from the naive $+165.4$ by the extra Lösschian representations of 49), $c_8 = -216$, $c_{10} = 225$, etc. Measured against the per-integer coefficient waveform of two disjoint generic subfamilies ($|D| \leq 60$ and $60 < |D| \leq 120$; 37 twists, $\sim 10^7$ samples, a_n computed by the verified lattice sieve), the parameter-free derived curve matches bin-by-bin within standard errors, with correlations $+0.959$ and $+0.996$ —each higher than the two subfamilies’ mutual correlation ($+0.949$): the derivation sits inside the data’s own noise. What remains is part two, the prime extraction: relating the per-integer density ρ to the per-prime waveform $\overline{M}$ through the multiplicative echo hierarchy; the measured prime-restricted column ($4\text{--}7\times$ larger than ρ with matching shape) is its target. The Grössencharacter murmuration density is derived and confirmed at the coefficient level.

4. THE REMAINING STEP, LOCATED

Closed-form murmuration densities for $\mathrm{GL}(1)$ over $\mathbb{Q}$ are established: Lee–Oliver–Pozdnyakov compute the densities for two families of Dirichlet characters, the real-character density exhibiting Zubrilina-type universality and interpolating the symplectic one-level phase transition [LOP]. The explicit-formula mechanism for quadratic families is given in [BEF], and Sarnak’s framework ties the density’s form to the conductor growth of the family [Sar, Zub]. The object measured here—the quadratic twist family of a fixed Grössencharacter of an imaginary quadratic field, conductor growth $N_D \asymp D^2$ —does not appear in these works. Its closed-form derivation is the application of the $\mathrm{GL}(1)$ template to a fully explicit input: the trace $a_p = \pi + \overline{\pi}$ is a lattice sum over $\mathbb{Z}[\omega]$, the weights are the closed forms of Proposition 2, and Poisson summation over the lattice has been carried out above; the coefficient-level density is derived in closed form and confirmed against the measured waveform. Observation 4 is the answer key any candidate density must reproduce.

Remark 7 (Scope). Proved and certified: Propositions 1–2, Corollary 3. Measured, with controls (universality across disjoint subfamilies, transform closure, deterministic reproduction): Observations 4–6. Not done: the closed-form evaluation of $\overline{M}$; it is posed with all ingredients supplied. Novelty of the measured object is asserted only as absence from the works cited, pending a full literature sweep. We claim nothing further.

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